GPU serving comparison and administrator change plan

Executive summary

The reference node has four H100 NVL GPUs arranged as two NVLink pairs. DeepInfer + DSpark currently serves through a LiteLLM gateway using an NCCL expert-exchange fallback. SGLang + DSpark is the retained alternate. DeepInfer is a separate inference engine, **not a vLLM backend**.

The matched direct-engine comparison found SGLang faster for one ordinary-text request, while DeepInfer sustained substantially more total output under the tested concurrent load. The detailed table appears once, including W1, W3 and W4. The predictable counting workload must not be treated as ordinary conversational performance.

The administrator request is to review the host configuration, trial a narrowly scoped change in an agreed maintenance window, and prove GPU copy integrity before trying DeepEP. No IOMMU/ACS/BIOS change, post-reboot recovery or post-change speed result is claimed.

Conditional expected outcome: for one fixed-length W1 request, an illustrative post-change model gives about **76-147 decode tokens/s**. Section 7 shows the assumptions and all derivable metrics. This is not measured, guaranteed or a forecast for 8/16 users.

SGLang single-user advantages depend on the workload: about 4.6% for W1 and 12.6% for W4. DeepInfer W4 total output at C16 is about 3.06x SGLang. Do not apply an older 7-9% range to this newer matrix.

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1 How the comparison was conducted

The same retained HTTP probe was used for both engines. It ran inside each engine pod against localhost:8000, bypassing LiteLLM, the context guard, DSH and the browser. The comparison measures these serving configurations; it does not isolate a single kernel or reproduce the upstream steady-state sweep.

ID	Task / exact prompt	Settings
W1	Write a detailed essay about the history of tea, with many sections.	512 output-token limit; ignore_eos=true
W3	List the integers from 1 through 40, separated by commas.	2048 output-token limit; normal EOS. Predictable task.
W4	Explain, in your own words, how a central bank sets interest rates and why.	512 output-token limit; normal EOS.

Every executed workload used temperature 0, streaming and server usage reporting. Clients after the first appended a variant number. The run did not explicitly force thinking off; reasoning tokens may be included in server counts.

Interaction and calculation

Read engine arguments, image identity, replicas and the sole operator/window; preserve the original probe and configuration.

For W1, W3 and W4, offer C=1,2,4,8,16 simultaneous requests, three repeats per cell. The active server cap is eight, so C16 includes queueing.

Record client SSE arrival times, server-reported usage, visible output and errors; calculate per-cohort metrics before taking the median of three repeat summaries.

Stop one four-GPU engine before starting the alternate; retain an independent restoration guard and the exact known-working configuration.

Restore DeepInfer and verify the Service endpoint and a real backend answer. Do not confuse that check with full authenticated DSH recovery.

Inventory: 3 workloads x 5 concurrency levels x 3 repeats = 45 cohorts and 279 completed request records per engine. This checks recorded completion/non-empty output, not comprehensive answer correctness.

Metric definitions: decode=(completion tokens-1)/(last-first non-empty SSE arrival). TPOT is 1000/unrounded decode, reduced per cohort then across repeats. Total output=sum(tokens)/(latest final token-earliest submission). Per-GPU output=total/4. TTFT=submission to first token of any kind. These are client stream metrics, not GPU step times or tail-latency percentiles.

Interruption: the first SGLang client segment reached its process alarm/time limit after 43 cohorts (exit 142; SIGALRM). Only the two incomplete W4/C16 cohorts were completed afterward on the same engine. Completed cohorts were not repeated. Extra interrupted attempts can exist outside the 279 accepted records; the resumed matrix is not one uninterrupted run.

No uniform cold-cache reset was applied, and direct background traffic was not comprehensively excluded. W3 can benefit disproportionately from predictable drafting; inclusion does not establish general conversational speed or exhaustive counting correctness. No new near-1M or GPU peer-integrity test is implied.

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/benchmarks/measured-results.csv

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/benchmarks/http_probe.py

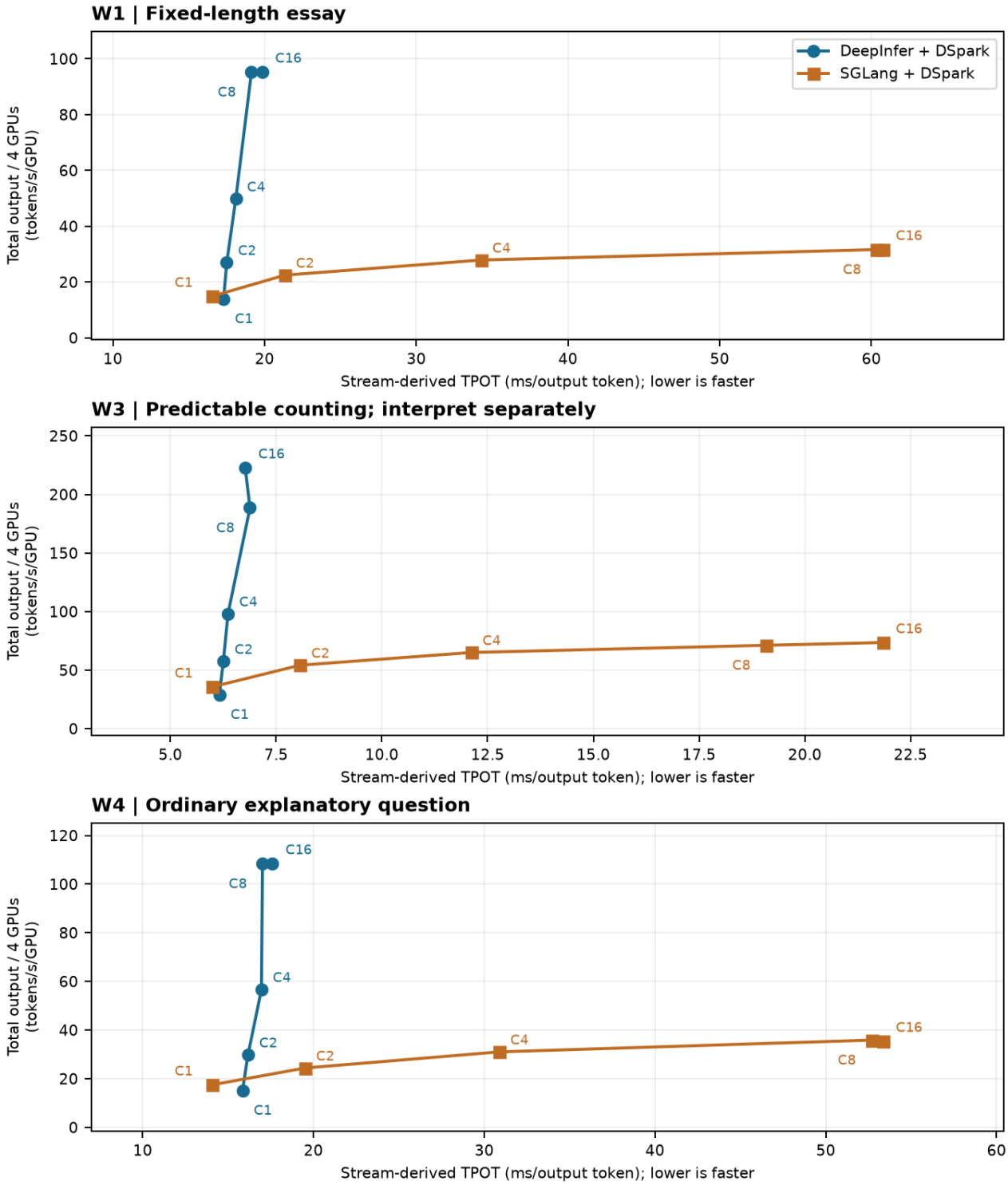
2 Measured results: W1, W3, W4

Four allocated GPUs in every row. C is offered concurrency, not requests per GPU; active capacity remains eight. Token ranges are actual reported input/output, not the 1M configured context ceiling. Each value is a median of three cohort summaries.

Workload / engine	C	Input / output tokens	Decode tok/s / request	TPOT ms/token	Total output tok/s	Output tok/s / GPU	TTFT ms
W1 / DeepInfer	1	18 / 512	57.8	17.30	56.0	14.01	294.9
W1 / DeepInfer	2	18-23 / 512	57.2	17.50	108.0	27.00	297.4
W1 / DeepInfer	4	18-23 / 512	55.3	18.10	199.2	49.80	299.7
W1 / DeepInfer	8	18-23 / 512	52.3	19.13	380.9	95.23	271.6
W1 / DeepInfer	16	18-23 / 512	50.4	19.83	381.2	95.30	4991.0
W1 / SGLang	1	18 / 512	60.4	16.54	59.6	14.90	130.7
W1 / SGLang	2	18-23 / 512	46.9	21.33	89.9	22.46	137.0
W1 / SGLang	4	18-23 / 512	29.2	34.27	111.5	27.89	290.9
W1 / SGLang	8	18-23 / 512	16.6	60.35	126.6	31.64	621.5
W1 / SGLang	16	18-23 / 512	16.5	60.75	126.7	31.68	16303.3
W3 / DeepInfer	1	18 / 119	162.4	6.16	116.5	29.13	294.7
W3 / DeepInfer	2	18-23 / 119	159.9	6.25	229.7	57.43	298.0
W3 / DeepInfer	4	18-23 / 119	157.2	6.36	392.3	98.07	349.7
W3 / DeepInfer	8	18-23 / 119	145.3	6.88	754.2	188.55	449.2
W3 / DeepInfer	16	18-23 / 119	147.7	6.77	891.3	222.81	818.8
W3 / SGLang	1	18 / 119	167.0	5.99	142.7	35.67	126.9
W3 / SGLang	2	18-23 / 119	124.0	8.07	216.7	54.18	146.2
W3 / SGLang	4	18-23 / 119	82.6	12.11	260.1	65.04	291.3
W3 / SGLang	8	18-23 / 119	52.4	19.08	284.8	71.19	629.1
W3 / SGLang	16	18-23 / 119	45.8	21.84	294.2	73.54	2001.9
W4 / DeepInfer	1	21 / 512	63.1	15.85	60.9	15.22	264.8
W4 / DeepInfer	2	21-26 / 512	61.9	16.17	119.2	29.80	262.6
W4 / DeepInfer	4	21-26 / 512	59.0	16.96	226.8	56.70	263.8
W4 / DeepInfer	8	21-26 / 473-512	58.8	17.02	433.5	108.36	266.1
W4 / DeepInfer	16	21-26 / 459-512	56.8	17.59	433.8	108.46	4501.4
W4 / SGLang	1	21 / 512	71.1	14.07	69.9	17.48	134.1
W4 / SGLang	2	21-26 / 447-512	51.3	19.50	97.6	24.39	136.1
W4 / SGLang	4	21-26 / 512	32.4	30.87	124.1	31.02	287.3
W4 / SGLang	8	21-26 / 210-512	19.0	52.67	143.5	35.86	624.5
W4 / SGLang	16	21-26 / 473-512	18.8	53.33	141.6	35.41	14711.9

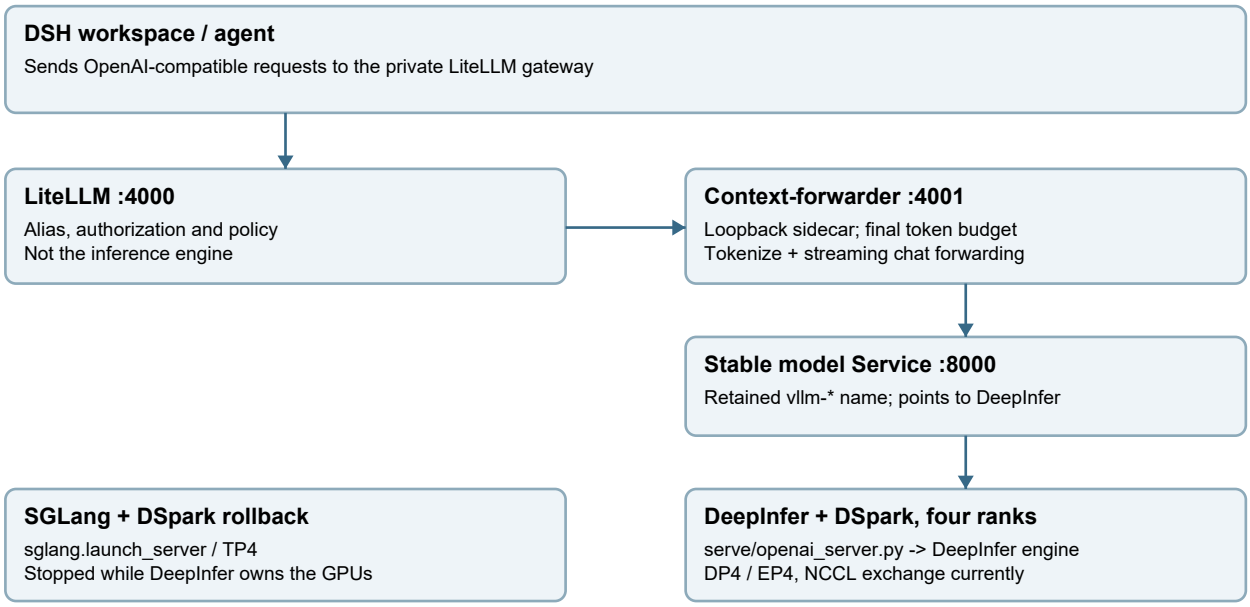
3 Throughput versus latency graph

Axes follow the useful throughput/latency presentation used by the LocalEngram work: output tokens/s per allocated GPU versus time per output token. Every plotted point is our own measured finite-cohort result; labels show offered C. No B200 result or conditional estimate is plotted as an H100 measurement.



Higher is more total work per allocated GPU; farther left is faster per request. Curves show the trade-off across offered load, not a new benchmark or an upstream Pareto-equivalence claim.

4 How LiteLLM reaches DeepInfer



Component	Actual role / distinction
DSH / LiteLLM	The workspace submits to the internal authenticated gateway. LiteLLM maps the model alias to a loopback context-forwarder. The openai/ adapter prefix means API compatibility, not an external OpenAI destination.
Context-forwarder	Enforces a final 1,048,576-token budget, 8,192 output ceiling and 256-token reserve. It uses the backend tokenizer and forwards streaming chat to the stable engine Service.
DeepInfer server	PID1 is python3 -u serve/openai_server.py. The source calls deepinfer._C.engine and DeepInfer sampling/runtime code. It does not launch vllm.entrypoints or sglang.launch_server.
Retained names / base image	A Service named vllm-* and a lmsysorg/sglang base-image digest do not prove which inference engine is running. Mounted code and the actual entry point matter.
Rollback engine	SGLang launches with python3 -m sglang.launch_server, TP4 and DSpark. It is stopped while DeepInfer owns the four GPUs; switching is operator-controlled, not automatic failover.

The humming MoE kernel package is associated with the vllm-project organization; using that library does not mean the vLLM serving engine runs this model. The current process still enters the DeepInfer server and engine directly.

The restored backend answered 2+2=4 from the existing LiteLLM pod through the model Service. That bypassed LiteLLM authentication and the full DSH route. No fresh gateway-overhead percentage is inferred. Older approximate 3% overhead observations require their own workload/window and are not merged into this direct-engine table.

https://github.com/skunpoj/DeepInfer/blob/b4df8657412de7ea8bab8b442c2b8c68f5c1068a/serve/openai_server.py

5 Current host and serving controls

Observed item	Value / meaning
Hardware / firmware	HPE ProLiant DL385 Gen11; BIOS 3.16; four H100 NVL, about 94 GiB each; NVIDIA driver 580.178.04. Individual BIOS controls remain unread.
OS / kernel	RHEL 9.8; kernel 5.14.0-687.52.1.el9_8.x86_64. The kernel build configuration was not inspected.
Topology	NVLink within GPU0-1 and GPU2-3; cross-pair paths are SYS across two NUMA domains. This is not a full four-GPU NVLink fabric.
Effective IOMMU	123 exposed groups report DMA-FQ; the four GPU groups also report DMA-FQ. No iommu=pt / iommu.passthrough=1 / iommu=off override appears in the observed boot arguments.
Interpretation	DMA-FQ is translated DMA with a flush queue; it must not be described as an observed translated-strict default. Read actual groups and kernel messages after any trial.
Device isolation	Four exposed GPU sriov_numvfs values are zero. This does not prove there are no VMs, VFIO users or other devices requiring DMA isolation.
Unknown controls	GPU upstream bridge ACS bits, current HPE workload preset, AMD I/O virtualization and DMA-remapping selections require administrator readback. Vendor default tables are not observed host state.
Present outcome	DeepInfer/NCCL serves; SGLang is retained stopped. No reboot, IOMMU/ACS/BIOS change, verified DeepEP fast path or post-change performance is reported.

Why a different setting may be needed

General-purpose firmware/kernel policies support isolation, virtualization and power-management needs; GPU peer traffic has different requirements. A container is not an independent IOMMU mode. Bare-metal operation does not remove the value of DMA isolation. Read the current controls and device users before choosing a trade-off; the observed correlation does not isolate IOMMU as the only cause.

Current serving configuration and source identity

Control	DeepInfer current	SGLang rollback
Engine / parallelism	DeepInfer OpenAI server; DP4 / EP4	SGLang launch_server; TP4
Context / active capacity	1,048,576; two sequences/rank, eight global	1,048,576; eight global
Memory / prefill	Target0.95; batch-token cap1024; prefill interval2	Static0.922; chunk1024; max prefill1024; prefill requests2; interval4
Speculation	NextN5; MTP varlen enabled	DSPARK block5; decode graph cap8
Expert / CP transport	EP=nccl; CP=nccl; GIN disabled; NVLS disabled; NCCL peer levelNVL	TP collectives; retain qualified scheduler/indexer overlays
Engram / cache	Host-UVA1; compact Engram0; hicache0GiB; YOCO keep0	Host table private/pinned; KV prefetch/drop-page-cache enabled; session radix cache;16SWA tails
Request / watchdog	thinking-mode chat; least-loaded routing; rank heartbeat120s	DeepSeek-v4.1 reasoning and tool parsers
What is not proved	Configured flags do not establish measured DSpark acceptance or full optimisation execution.	Rollback specification is not simultaneous GPU service or a host-reboot recovery receipt.

Selected running-source comparison against public fork commit b4df8657 found **264 matching files and one different file: serve/prompt_tokenizer.py**. Some C++/build/native files were absent in the runtime filesystem. Compiled libraries, vendor submodules, weights and extra untracked files were not established by this comparison. The fork is available, but exact clean-checkout-to-running-binary reproduction remains open.

The three public profiles and existing renderer are reference code. Private site overlays and the current Azure operational build still exist separately; the CI/CD source cutover has not been completed by publishing these profiles.

6 Administrator procedure: read, change, verify

A. Read-only host checks and recovery preparation

```
uname -r
cat /proc/cmdline
cat /sys/kernel/iommu_groups/*/type | sort | uniq -c
nvidia-smi topo -m
lspci -t
lspci -vv
grubby --default-kernel
grubby --info=ALL
```

Record every GPU upstream bridge, ACS controls, driver binding, active VF and VFIO/VM/device-assignment user. Save exact boot entries, private workload specifications, images, mounted source/binary hashes, storage bindings and routing. Confirm console recovery and a known-working boot entry. Assign a sole operator and a restoration deadline.

B. First controlled OS trial

In an explicitly agreed maintenance window, the administrator may trial iommu=pt on a selected kernel entry after reviewing isolation impact. Preserve a recovery entry. Do not blindly change all installed kernels or edit every boot configuration file together.

```
# Example only: administrator selects the actual trial entry
grubby --update-kernel=/boot/vmlinuz-TRIAL --args="iommu=pt"
# Inverse for that same trial entry, if rollback is required
grubby --update-kernel=/boot/vmlinuz-TRIAL --remove-args="iommu=pt"
```

On a BLS-based RHEL installation, inspect /boot/loader/entries/ and the actual kernel-install/managed boot configuration. Verify the effective command line and GPU domain types after reboot. Do not assume grubby automatically updates /etc/default/grub and /etc/kernel/cmdline, or that editing those files proves persistence. Retest persistence after the next managed kernel update.

C. Integrity and conditional escalation

Check all required peer-copy directions with multiple sizes and seeded nonzero patterns; verify destination bytes. A P2P capability flag or an implausible bandwidth figure without correctness is not a pass. Then test DeepEP dispatch/combine and model correctness. Only if integrity still fails, review the exact ACS bridge bits or firmware profile with the hardware owner. No system-wide ACS clearing is proposed.

Cordon does not stop running pods. Coordinate eviction, local storage, disruption budgets and GPU ownership. Host boot recovery requires console/boot-entry recovery; a Kubernetes process watchdog cannot repair a failed host boot.

Administrator procedure: exact setup locations and restoration

Where the three configurations are kept

One public setup home: skunpoj/dsh_setup/gpu-stack. The following files separate the retained SGLang alternate, current DeepInfer fallback and proposed DeepEP trial. The actual private Azure operational repository remains GPU_server until cutover. These public references do not claim that cutover has happened.

SGLang + DSpark rollback

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/profiles/sglang-dspark-rollback.json

DeepInfer + DSpark, current NCCL fallback

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/profiles/deepinfer-dspark-nccl-current.json

DeepInfer + DSpark, proposed after-change trial

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/profiles/deepinfer-dspark-deepep-proposed.json

Code and full change/recovery runbook

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/profiles/inspect_profile.py

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/runbooks/gpu-peer-access.md

The inspector validates and prints an argument/environment plan; it never runs an engine or applies cluster resources. The proposed profile changes only expert transport from nccl to deepep. CP stays nccl, GIN stays disabled and Engram stays host-UVA; unrelated settings are held fixed for a controlled trial.

The SGLang Kubernetes reference and renderer preserve its gate/overlay design. The DeepInfer engine itself is in the separate upstream fork; a single setup repository is not the same thing as a single combined engine source tree.

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/templates/model.yaml

https://github.com/skunpoj/dsh_setup/blob/main/gpu-stack/render.py

https://github.com/skunpoj/DeepInfer/tree/b4df8657412de7ea8bab8b442c2b8c68f5c1068a

Restoration and acceptance

On boot failure, missing devices, byte mismatch, model failure or deadline expiry, restore the saved boot entry and serving specification. Do not run both four-GPU engines simultaneously. Read back source/image/Service identity, prove answer correctness, verify the full authenticated LiteLLM/DSH path and confirm unchanged workspace/history/storage. A 1500-second operational watchdog is not a demonstrated host-reboot RTO.

7 Expected outcome: a conditional model, not a measurement

Plain meaning: this row asks what one W1 request might achieve if the host fix and DeepEP trial succeed, reduce the assumed communication cost, and leave the other assumptions unchanged. It does not say that the future system has already passed qualification.

W1 / one request	Decode tok/s / request	TPOT ms/token	Total output tok/s	Output tok/s / GPU	TTFT ms
Current measured baseline	57.8	17.30	56.0	14.01	294.9
After successful host + DeepEP trial: conditional calculation	76.3-146.7	6.82-13.11	73.2-135.5	18.31-33.89	294.9 (held constant)

Assumptions and derivation

The retained profiler-based scenario starts with a 33 ms step and attributes 29 ms to the expert path. Assuming a replacement path of 9-21 ms gives a 13-25 ms step. Holding accepted output per step constant at the current W1 decode rate multiplied by 0.033 gives the decode range above (about 1.32-2.54x). These are model inputs, not isolated causal measurements or validated hardware limits.

TPOT=1000/decode. For this one-request, 512-output-token scenario only, total output=512/(0.2949+511/decode), and per-GPU output=total/4. TTFT is explicitly held at the observed 294.9 ms; no improvement is predicted. Input 18 tokens, output 512 tokens, offered C=1, allocated GPUs=4 and global active cap 8 are fixed assumptions, not estimated gains.

The baseline column uses independently reduced observed metrics, so it need not equal a formula applied to separately rounded medians. The conditional row uses a single consistent arithmetic model. Different input/output lengths, acceptance, memory pressure or TTFT would change it.

Cannot responsibly forecast from these inputs: C8/C16 total output, first-token improvement, P95/P99 latency, full-1M performance, prefix-cache gains or reboot recovery time. No numeric estimates are invented for them. Re-measure after integrity, dispatch/combine and source/binary gates pass.

https://github.com/skunpoj/dsh_setup/blob/main/docs/gpu-platform/conditional-scenario.json

8 What vendor guidance supports, and what it does not

Claim in supplied material	Reviewed wording for this report
NVIDIA/HPE officially support every proposed change	Their documents describe relevant P2P limitations and available firmware controls. That is not certification of this exact host, engine, passthrough policy or a guaranteed fix.
RHEL9 is translated-strict on this host	Not established. Actual exposed groups report DMA-FQ. The installed kernel configuration and all device policies were not read.
Passthrough preserves the same device isolation	Do not assume this. Identity-mapped device DMA does not provide the same per-device address translation boundary. The administrator must review device ownership and isolation requirements.
iommu=pt equals iommu.passthrough=1 everywhere	They express related passthrough intent in documented kernel interfaces. Installed enterprise-kernel support and interaction with other parameters/device policies must be confirmed.
Linux enables every listed ACS bit whenever IOMMU is present	Too broad. Bridge capability, firmware and kernel policy vary. Read ACSCtl on each relevant bridge; IOMMU presence alone does not prove the bits or the routing.
HPE Graphic Processing preset is the required fix	A documented bundled preset, not an observed current setting or a mandatory change. The processor-specific table can disable I/O virtualization/SR-IOV and alter fabric/power policy; review each affected control first.
grubby persists every listed file automatically	Do not generalize. Inspect BLS and the host boot-management chain, then verify effective arguments after boot and future managed kernel updates.
Zero GPU VFs proves no VM/VFIO dependency	False inference. It describes those GPU functions only. Other devices, hypervisors and device-assignment users need review.

The first proposed experiment changes one OS-level factor. A scoped ACS software trial or firmware change is a later, separately reviewed escalation. Neither vendor guidance nor the current correlation proves the source of the destination-data mismatch.

9 GitHub references and their relation to this setup

DeepInfer PR47 - Design RFC for single-server SM90/H20 support; not a delivered H100 implementation.

https://github.com/deepseek-ai/DeepInfer/pull/47

DeepInfer PR92 - Community SM90/Hopper foundation. Its H20 benchmark is not this H100 finite-cohort run.

https://github.com/deepseek-ai/DeepInfer/pull/92

DeepInfer PR147 - OpenAI-compatible HTTP serving proposal; related to our DeepInfer entry point, not vLLM.

https://github.com/deepseek-ai/DeepInfer/pull/147

DeepInfer PR200 - Host-pinned UVA Engram proposal; related mechanism to the current host-UVA flag.

https://github.com/deepseek-ai/DeepInfer/pull/200

DeepInfer PR144 - Host prefix-cache proposal; not enabled by current hicache0 / YOCO0 values.

https://github.com/deepseek-ai/DeepInfer/pull/144

DeepInfer PR246 - Opt-in GPU LocalEngram on four B200 GPUs/full NVLink. Inspired the axes, not our data or a compatible H100 claim.

https://github.com/deepseek-ai/DeepInfer/pull/246

deepinfer-hopper community repository - Related Hopper adaptation with separate pins/qualification; not our deployed source.

https://github.com/kabishou11/deepinfer-hopper

DeepEP - Expert dispatch/combine library for the proposed trial; presence is not a peer-integrity pass.

https://github.com/deepseek-ai/DeepEP

humming kernels - MoE kernels under vllm-project; library reuse is distinct from running the vLLM serving engine.

https://github.com/vllm-project/humming

Official sweep and plot - Steady-state step/acceptance accounting differs from our client-observed HTTP cohorts.

https://github.com/deepseek-ai/DeepInfer/blob/main/examples/sweep/plot_pareto.py

Our DeepInfer discussion221 - Deployment observations and benchmark discussion; current report distinguishes measured facts and proposed changes.

https://github.com/deepseek-ai/DeepInfer/discussions/221

Our Harness discussion5682 - Gateway/multi-tenancy integration proposal. Engine benchmarks do not qualify every product feature.

https://github.com/deepseek-ai/deepseek-harness/discussions/5682

These are the selected related references requested for this review, not an exhaustive inventory of every upstream PR or fork. Official sweep, SM90 request benchmark and LocalEngram hardware/workloads differ; matching units alone does not make experiments equivalent.

Reference links for the administrator

NVIDIA CUDA

Bare-metal IOMMU/PCIe P2P limitation; does not establish this host's isolated cause.

https://docs.nvidia.com/cuda/cuda-programming-guide/03-advanced/multi-gpu-systems.html#host-iommu-hardware-pci-access-control-services-and-vms

NVIDIA NCCL troubleshooting

Integrity checks, translated GPU-path caveat and ACS review.

https://docs.nvidia.com/deeplearning/nccl/user-guide/docs/troubleshooting/gpu_troubleshooting.html

Linux kernel parameters

Definitions and platform-specific parameter scope.

https://www.kernel.org/doc/html/latest/admin-guide/kernel-parameters.html

RHEL kernel arguments

Boot-management procedure; actual host layout and persistence still require readback.

https://docs.redhat.com/en/documentation/red_hat_enterprise_linux/9/html/managing_monitoring_and_updating_the_kernel/configuring-kernel-command-line-parameters_managing-monitoring-and-updating-the-kernel

HPE profile dependencies

Processor-generation-specific preset table; use the matching host generation/firmware.

https://support.hpe.com/hpesc/public/docDisplay?docId=sd00003788en_us&page=GUID-0046C58A-060D-4089-9B21-9AFEC16D7F74.html

Kubernetes drain

Cordon/eviction/disruption and storage implications.

https://kubernetes.io/docs/tasks/administer-cluster/safely-drain-node/

LiteLLM API adapter

Adapter selection and a configurable OpenAI-compatible target; not proof of external routing.

https://docs.litellm.ai/docs/providers/openai_compatible

Evidence and remaining acceptance

The retained evidence includes the original probe, both benchmark segments, configuration readbacks, startup gates, restoration Service answer, selected engine-source hashes and focused earlier transcript passages. No new model call, speed run, host change or paid service was used to produce this revision. Link availability and a positive readiness flag are not runtime acceptance.

Open items remain exact engine/binary provenance, host/ACS/firmware confirmation, post-change GPU correctness, controlled DeepEP performance, host reboot recovery and full authenticated user-route acceptance. Operational identifiers, internal service addresses, private evidence and historical originals are retained privately; this public edition is sanitized.